02 · Case 2: the aftereffects of lymph node removal

The words of this track, as it says them. Bold brown is the examiner.

Exam case 2. The aftereffects of lymph node removal.

Aleki and Vicky visit their friend Mary, who had a right mastectomy for cancer a fortnight ago. Mary mentions the cancer had spread to the lymph nodes, and some axillary lymph nodes were also removed. She complains of swelling in her right arm since the surgery. Her friend, who is good with mirimiri and massage, offers to massage her arm, which helps bring the swelling down.

Three questions, seven marks. It's the shortest case in the exam, so these are the easiest marks to bank.

Three marks. What are the major functions of the lymphatic system?

Three functions, a mark each. The hook: digest, defend, drain.

Digestive. Lacteals in the gut absorb dietary fat, and carry it away for processing.

Immunological. The immune response. The thymus, where T cells mature. The bone marrow, where B cells mature, and B cells make antibodies. And the lymph nodes, where lymphocytes and other immune cells meet and coordinate an immune response.

Fluid balance. It drains away the excess tissue fluid, the interstitial fluid, monitors it for antigens on the way, and returns it to the venous system.

Two marks. What is the role of the axillary lymph nodes in the spread of cancer? Explain.

The spread of cancer is called metastasis. If a cancerous cell breaks away from its tumour and gets into the lymphatic system, it can potentially reach any part of the body.

The breast drains to the axillary nodes, in the armpit, so they're checked for cancer cells. The first node in line, closest to the breast, is the sentinel node. Cancer cells there show the cancer has spread. No cancer cells there gives some confidence that it's still limited to the breast.

Two marks. Why did Mary develop swelling in her right arm after the mastectomy? Think about fluid dynamics, and what keeps fluid in the tissue spaces.

When the lymph nodes are removed, the fluid that normally drains from the arm through those nodes is blocked. Normally, tissue fluid drains through the lymphatic vessels, and back into the venous system at the subclavian vein. Now it has nowhere to go. The lymphatic vessels fill until they can't drain the tissue, and fluid stays in the interstitial spaces. That swelling is lymphoedema.

Her formative test turns this case into drop-downs, and the keys are worth knowing word for word. The role of the axillary nodes in the spread of cancer: metastasis. No cancer cells in

them shows the cancer is limited to the breast. And Mary's swelling is a sign that the lymph drainage is hampered by the surgery.

Once more. Case 2.

The lymphatic system has three jobs. Digestive: lacteals absorb dietary fat. Immunological: the thymus, the bone marrow and the lymph nodes. And fluid balance: draining excess tissue fluid back to the venous system. Cancer spreading is metastasis, and cells in the lymph can reach any part of the body. The axillary nodes are checked, the sentinel node first: cancer there means it has spread; none means it's likely limited to the breast. Removing the nodes blocks the arm's drainage to the subclavian vein, so fluid stays in the tissue: lymphoedema.